

SOC Investigation: Finance Workstation Compromise - Process Hollowing (PID 3880)

Incident Overview

Alert: Generic Suspicious Network Connection - Finance Dept Workstation

Antivirus Status: No malicious files detected on disk

Evidence Source: Volatility memory dump analysis

Investigator: Oyewole Samad

Executive Summary

A compromised `svchost.exe` (PID 3880) was identified on the Finance workstation. The process was spawned by `notepad.exe` (abnormal parent-child relationship), contained injected executable code in `PAGE_EXECUTE_READWRITE` memory with MZ magic bytes, and maintained an active TCP connection to external IP `185.112.55.20:443`. This is consistent with **Process Hollowing (T1055.012)**.

1. Compromised Process Identification

```
processTree.txt
1 Name PID PPID Threads Session Time
2 System 4 0 124 - 2026-03-10 14:00:01
3 smss.exe 340 4 2 - 2026-03-10 14:00:02
4 wininit.exe 456 340 3 0 2026-03-10 14:00:05
5 services.exe 512 456 12 0 2026-03-10 14:00:06
6 svchost.exe 720 512 24 0 2026-03-10 14:00:07
7 svchost.exe 884 512 18 0 2026-03-10 14:00:07
8 lsass.exe 520 456 8 0 2026-03-10 14:00:06
9 winlogon.exe 580 340 4 1 2026-03-10 14:00:06
10 explorer.exe 1840 580 72 1 2026-03-10 14:05:22
11 chrome.exe 2112 1840 34 1 2026-03-10 14:10:15
12 notepad.exe 2550 1840 2 1 2026-03-10 14:45:01
13 svchost.exe 3880 2550 5 1 2026-03-10 14:46:12
14
15
16 PROCESS TREE ANALYSIS
17
18 System (PID 4)
19   smss.exe (PID 340)
20     wininit.exe (PID 456)
21       services.exe (PID 512)
22         svchost.exe (PID 720)
23         svchost.exe (PID 884)
24       lsass.exe (PID 520)
25     winlogon.exe (PID 580)
26       explorer.exe (PID 1840)
27         chrome.exe (PID 2112)
28         notepad.exe (PID 2550)
29           svchost.exe (PID 3880)
30
```

The compromised process identified is:

- **Process Name:** `svchost.exe`
- **PID:** 3880

This process deviates from expected system behavior and exhibits both malicious memory characteristics and suspicious network activity. The `svchost.exe` should be spawned by `services.exe`. Therefore, this is suspicious and requires further investigation because it is an illegitimate process relationship.

2. Parent-Child Analysis

- **Parent Process:** `notepad.exe`
- **Parent PID:** 2550

This relationship is highly suspicious because:

In a standard Windows environment, `svchost.exe` must be launched by `services.exe` (PID 512). A `svchost.exe` instance spawned by `notepad.exe` indicates

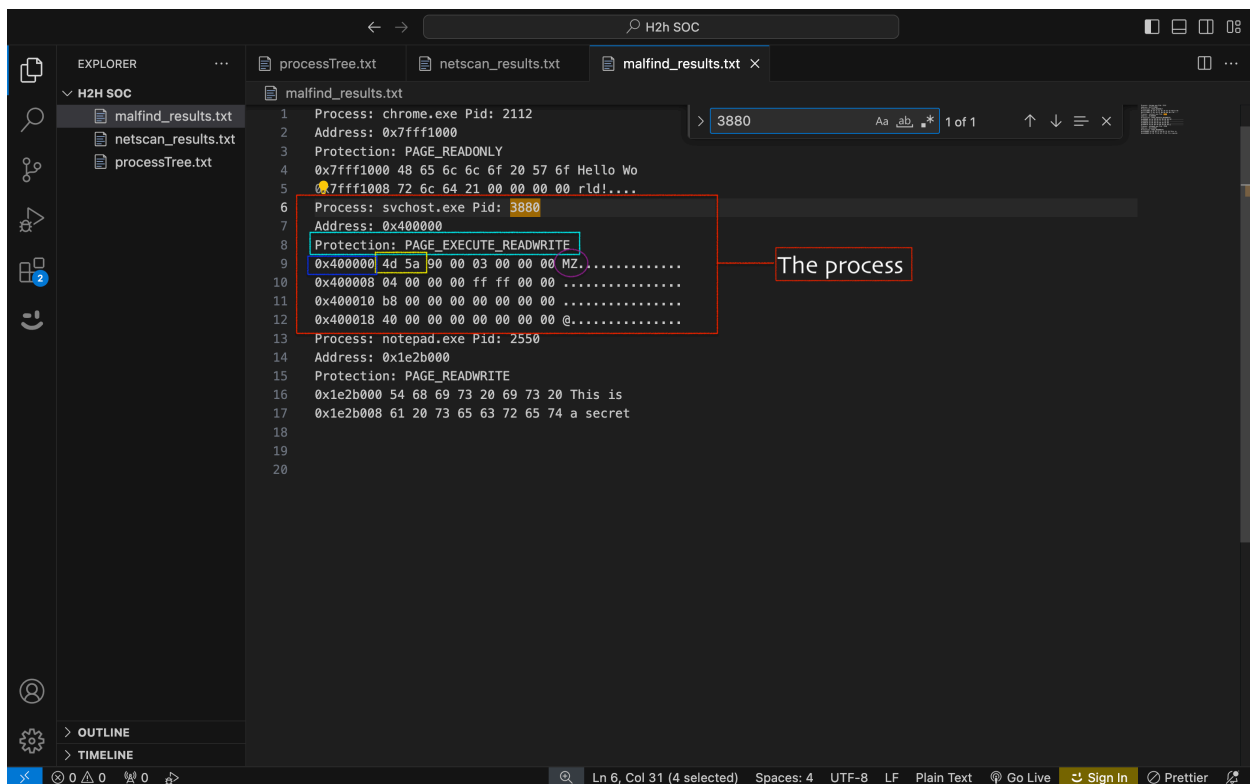
process masquerading or code injection from a user application. This represents a clear architectural violation.

```
1 Name PID PPID Threads Session Time
2 System 4 0 124 - 2026-03-10 14:00:01
3 smss.exe 340 4 2 - 2026-03-10 14:00:02
4 wininit.exe 456 340 3 0 2026-03-10 14:00:05
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8 lsass.exe 520 456 8 0 2026-03-10 14:00:06
9 winlogon.exe 580 340 4 1 2026-03-10 14:00:06
10 explorer.exe 1840 580 72 1 2026-03-10 14:05:22
11 chrome.exe 2112 1840 34 1 2026-03-10 14:10:15
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16 PROCESS TREE ANALYSIS
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19   └─ smss.exe (PID 340)
20       └─ wininit.exe (PID 456)
21           └─ services.exe (PID 512)
22               └─ svchost.exe (PID 720)
23                   └─ svchost.exe (PID 884)
24           └─ lsass.exe (PID 520)
25       └─ winlogon.exe (PID 580)
26           └─ explorer.exe (PID 1840)
27               └─ chrome.exe (PID 2112)
28                   └─ notepad.exe (PID 2550)
29                       └─ svchost.exe (PID 3880)
```

normal svchost.exe under services.exe

malicious svchost (3880) under notepad.exe

3. Memory Injection Evidence (malfind)



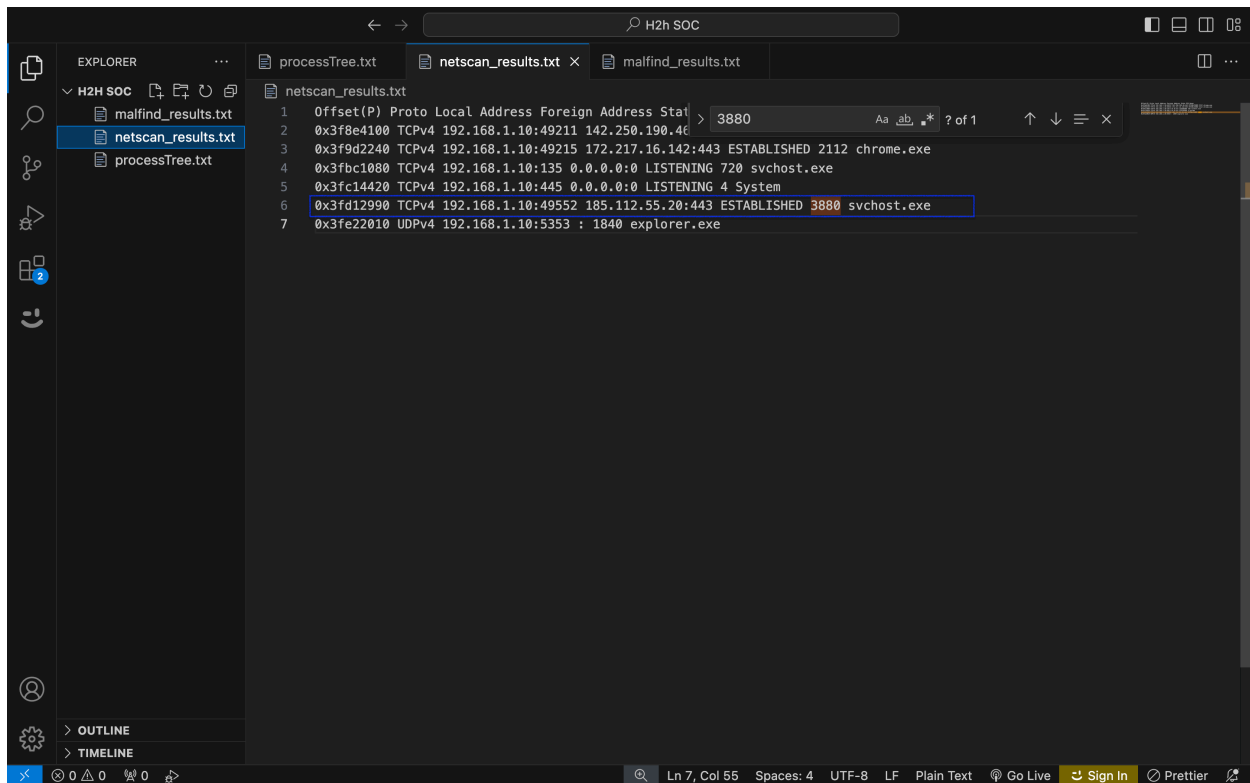
Using PID 3880 to filter the malfind results, we identified the following memory artifacts:

- **Process:** `svchost.exe` (PID 3880)
- **Memory Address:** `0x400000`
- **Memory Protection Flags:** `PAGE_EXECUTE_READWRITE`
- **Magic Bytes:** `4d 5a` (MZ Header)

These findings confirm that executable code has been injected into the process memory. The presence of an MZ header indicates that a Portable Executable (PE) file is running directly from memory.

- `PAGE_EXECUTE_READWRITE` is rare for legitimate system processes (normal is `PAGE_EXECUTE_READ`)
- `4d 5a` (MZ) is the Windows executable signature — this code was never loaded from disk
- Combined, this confirms **code injection**

4. C2 Network Communication



```
1 Offset(P) Proto Local Address Foreign Address State ... 3880
2 0x3f8e4100 TCPv4 192.168.1.10:49211 142.250.190.46
3 0x3f9d2240 TCPv4 192.168.1.10:49215 172.217.16.142:443 ESTABLISHED 2112 chrome.exe
4 0x3fbc1080 TCPv4 192.168.1.10:135 0.0.0.0:0 LISTENING 720 svchost.exe
5 0x3fc14420 TCPv4 192.168.1.10:445 0.0.0.0:0 LISTENING 4 System
6 0x3fd12990 TCPv4 192.168.1.10:49552 185.112.55.20:443 ESTABLISHED 3880 svchost.exe
7 0x3fe22010 UDPv4 192.168.1.10:5353 : 1840 explorer.exe
```

Using PID 3880 to filter the network scan results, we identified the following C2 communication:

- **Destination IP Address:** 185.112.55.20
- **Port:** 443
- **Protocol:** TCPv4
- **Connection State:** ESTABLISHED

Why this is malicious:

- The IP does NOT belong to Microsoft, Google, Cloudflare, or any legitimate CDN
- Legitimate `svchost.exe` does not maintain active HTTPS connections to random hosting providers
- Port 443 is commonly used by attackers to blend in with normal web traffic

5. Attack Classification

Based on the combined evidence:

- Abnormal parent-child relationship
- Presence of injected executable code in memory
- Active outbound network communication

👉 The attack is classified as:

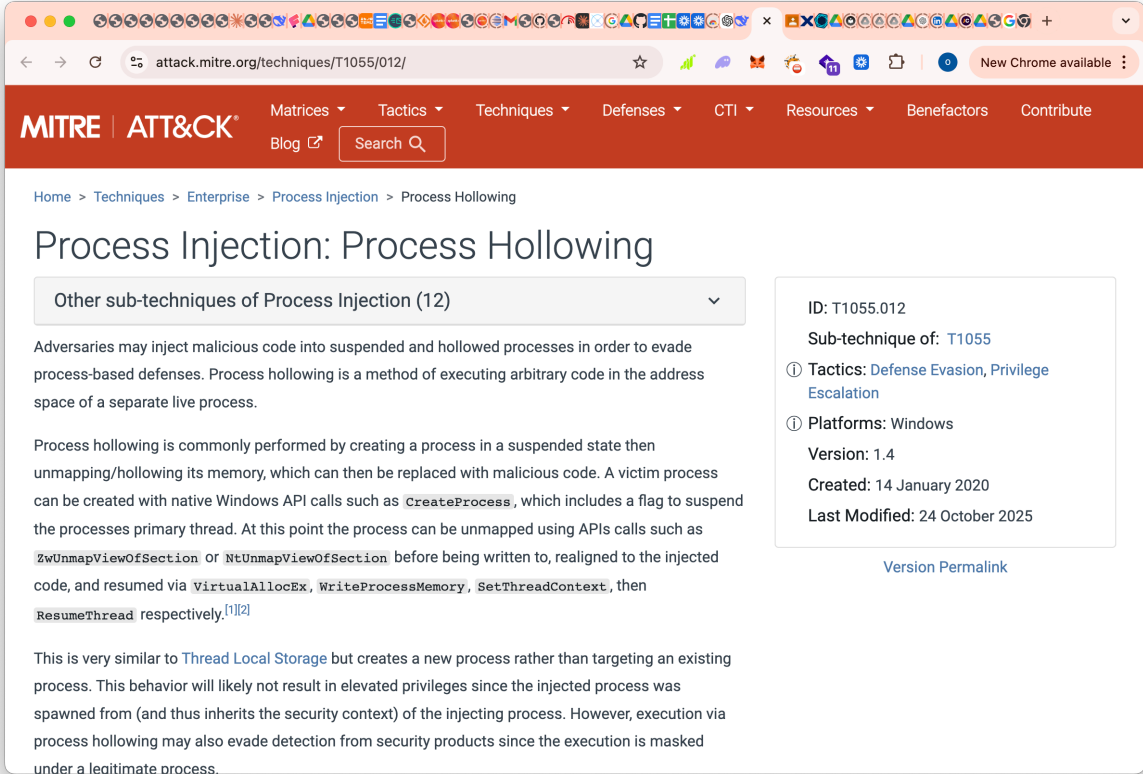
Process Hollowing (T1055.012)

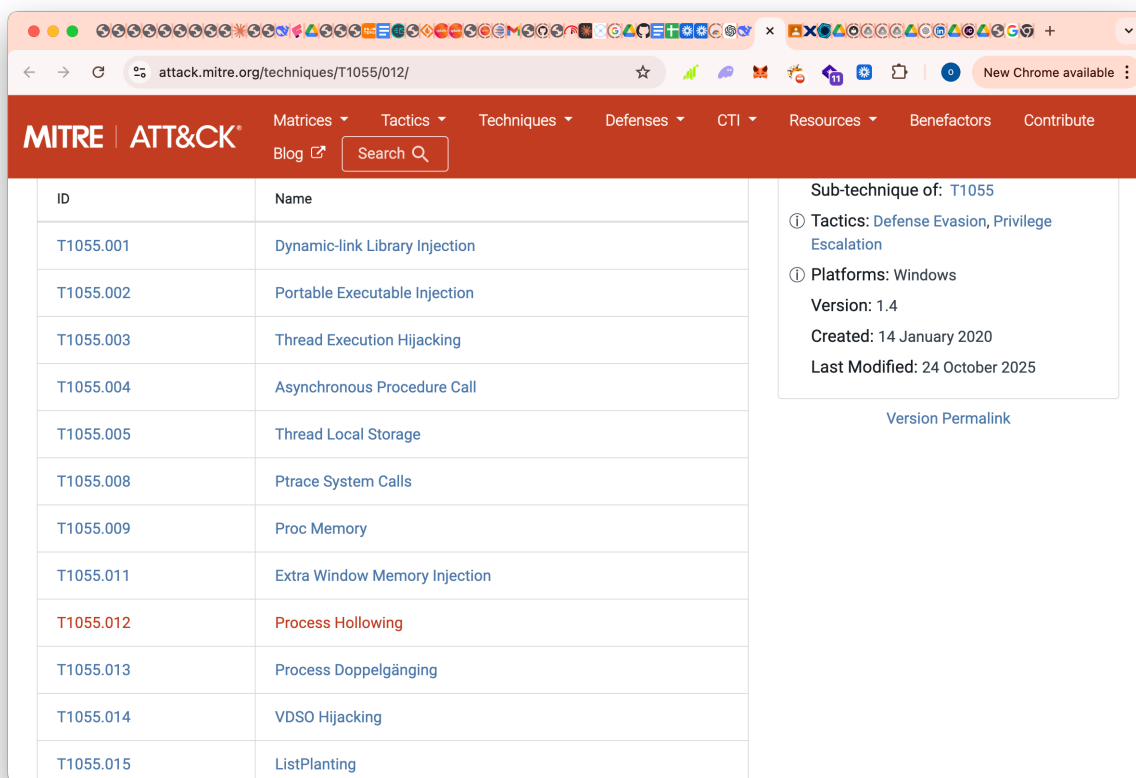
MITRE Tactic : Defense Evasion, Execution

This technique allows attackers to disguise malicious activity within legitimate system processes while executing entirely in memory.

Justification:

Evidence	Finding	Indicator
Process Tree	svchost.exe spawned by notepad.exe	Wrong parent for system process
Memory Analysis	PAGE_EXECUTE_READWRITE + MZ bytes at 0x400000	Injected executable code
Network Analysis	TCP connection to 185.112.55.20:443	Active C2 beaconing
Disk	No malicious file detected	Code is memory-resident only





Recommendations

Priority	Action
High	Isolate Finance workstation from network immediately
High	Block outbound traffic to 185.112.55.20 at firewall
Medium	Dump full memory of PID 3880 for further analysis (strings, YARA)
Medium	Check for persistence (scheduled tasks, run keys, WMI)
Low	Review other processes spawned by notepad.exe across the fleet

IOC Summary (For Threat Intelligence)

Indicator	Type	Notes
185.112.55.20	IP Address	C2 Server (port 443)
svchost.exe (PID 3880)	Process Name	Masquerading system process
notepad.exe → svchost.exe	Parent-Child Relationship	Abnormal process hierarchy
0x400000 + MZ bytes (4D 5A)	Memory Pattern	Process Hollowing signature
PAGE_EXECUTE_READWRITE	Memory Protection	Code injection indicator